

Qatar-1b

R-2624

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190612 · created 2026-08-28T23:44:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458646.7272 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.05
Transit depth (Rp/Rs)^2	0.63 +/- 0.8 [%]
Orbital Inclination (inc)	85.41 +/- 2.88
Airmass coefficient 1 (a1)	1.19 +/- 0.048
Airmass coefficient 2 (a2)	-0.094 +/- 0.024
Scatter in the residuals of the lightcurve fit is	4.05 %
Best Comparison Star	#1 - [382, 422]
Optimal Method	PSF photometry
Transit Duration (day)	0.065 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.079 ± 0.05 vs 0.1463 (deviation 1.35σ)
Duration fit vs published	0.065 d vs 0.0692 d (deviation 0.25σ)
Mid-time O–C	3.7 min
Depth SNR	1.6
Residual scatter	4.05 %

Light curve

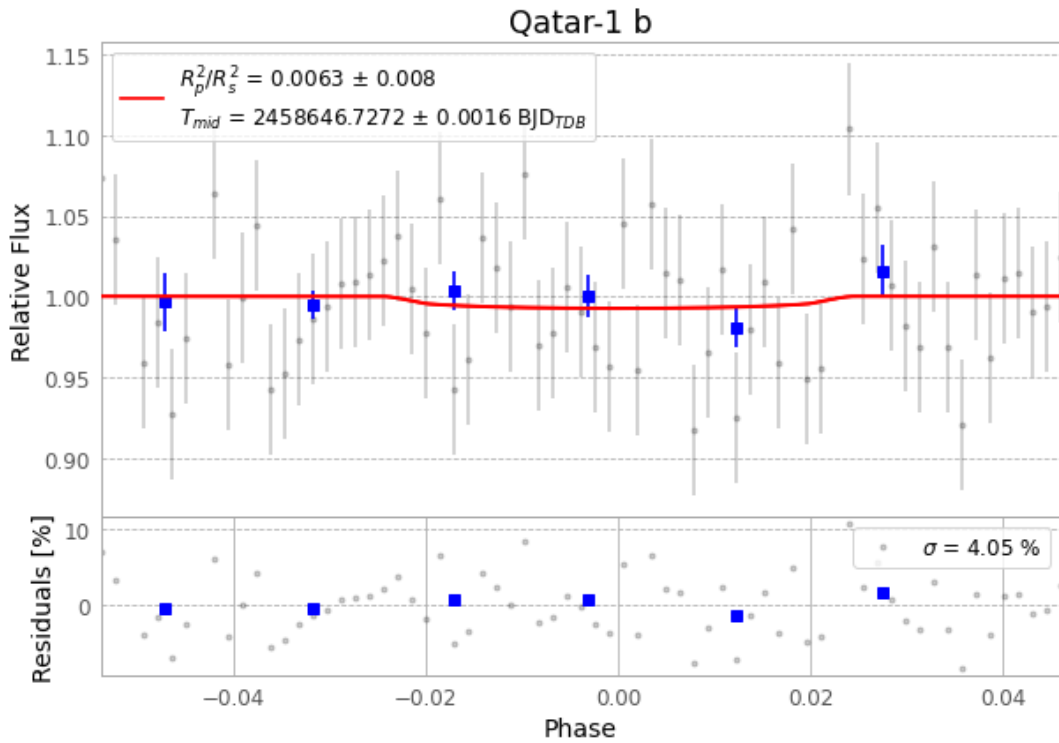


Figure 1 — FinalLightCurve_Qatar-1 b_2019-06-11.png (SHA-256 89dc371dd8bc0222...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 140], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d70e5c8873558c2b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), Qatar-1190612033612.FITS → Qatar-1190612070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190612.log (33c511b21274...), FinalLightCurve_Qatar-1 b_2019-06-11.png (89dc371dd8bc...), FinalLightCurve_Qatar-1 b_2019-06-11.csv (9ceb26befcc9...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:44Z.